

Solution Requirements

Date	02-July-2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Interface Portal	System presents an interactive home layout console that serves as the entry checkpoint to access the risk assessment terminal.	High
2	Telemetry Data Capture	Multi-column grid interface contains explicit input fields and categorical dropdown select elements to accept 11 specific demographic and financial attributes.	High
3	Browser-Side Validation	Frontend framework natively implements form field restrictions to prevent the submission of empty fields.	High
4	Input Stream Sanitization	Backend application route handles incoming request form structures by enforcing explicit floating-point data type casting (<code>float()</code>).	High
5	Feature Optimization Layer	Preprocessed numeric data arrays are scaled via a pre-fit <code>scale1.pkl</code> <code>StandardScaler</code> binary to establish	Medium

		standardized vector distributions before model scoring.	
6	Predictive Risk Scoring	Underwriting micro-controller interfaces with a pre-trained Random Forest model binary (<code>rdf.pkl</code>) to execute multi-tree split probability scoring on the clean feature matrix.	High
7	Dynamic Result Generation	Web layout utilizes conditional template compilation rules (<code>{% if ... %}</code>) to dynamically toggle approved/disapproved design views depending on classification outcomes.	High
8	Session Re-initialization	Interface results view provides active action controls allowing the evaluator to re-initialize form fields for fresh assessment sweeps or return to the main landing portal console.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Predict loan eligibility quickly.	Prediction within average 0.2 seconds
2	Scalability	Support multiple simultaneous users.	Zero thread blockages or connection stalls up to 50 virtual user loads.
3	Security & Input Integrity	Backend routing boundaries must neutralize malformed data payloads or string injections before vector calculations.	100% rejection of untyped alphanumeric data via explicit backend float casting exceptions.
4	Reliability & Availability	System available whenever accessed.	99.9% uptime
5	Usability & Accessibility	Responsive, user-friendly interface.	Responsive web application

6	Other	Maintainable and easy to update ML models.	Modular architecture
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